

Java Method & Constructor Overloading Tasks

1. METHOD OVERLOADING

1. printBill() – Shopping Bill

Scenario: Create a Bill class with overloaded printBill() methods to display product name; product name and quantity; product name, quantity and price.

Input:

```
Enter product name: Pen
Enter quantity: 5
Enter price: 20
```

Output:

```
Product: Pen
Product: Pen, Quantity: 5
Product: Pen, Quantity: 5, Price: 20
```

2. showTime() – Time Display

Scenario: Create a Time class with overloaded showTime() methods to display hours; hours and minutes; hours, minutes and seconds.

Input:

```
Enter hours: 10
Enter minutes: 30
Enter seconds: 45
```

Output:

```
Time: 10
Time: 10:30
Time: 10:30:45
```

3. calculate() – Salary Calculation

Scenario: Create a Salary class with overloaded calculate() methods to calculate salary using basic salary; basic salary and bonus; basic salary, bonus and allowance.

Input:

```
Enter basic salary: 25000
Enter bonus: 3000
Enter allowance: 2000
```

Output:

```
Basic Salary: 25000
Salary with Bonus: 28000
Total Salary: 30000
```

2. CONSTRUCTOR OVERLOADING

1. Laptop Configuration

Scenario: Create a Laptop class with overloaded constructors to initialize brand; brand and RAM; brand, RAM and storage.

Input:

```
Enter brand: Dell
Enter RAM: 16
Enter storage: 512
```

Output:

```
Brand: Dell

Brand: Dell
RAM: 16 GB

Brand: Dell
RAM: 16 GB
Storage: 512 GB
```

2. Movie Details

Scenario: Create a Movie class with overloaded constructors to initialize movie name; movie name and language; movie name, language and rating.

Input:

```
Enter movie name: Interstellar
Enter language: English
Enter rating: 9.0
```

Output:

```
Movie: Interstellar

Movie: Interstellar
Language: English

Movie: Interstellar
Language: English
Rating: 9.0
```

3. Ticket Booking

Scenario: Create a Ticket class with overloaded constructors to initialize passenger name; passenger name and seat number; passenger name, seat number and ticket price.

Input:

```
Enter passenger name: Arun
Enter seat number: A15
Enter ticket price: 850
```

Output:

```
Passenger: Arun

Passenger: Arun
Seat: A15

Passenger: Arun
```

Seat: A15

Ticket Price: 850